
The Edit Functions

About this Chapter

This chapter focuses on CADD's editing module. The edit functions make CADD a very powerful drawing tool and contribute to significant timesaving in design and drafting. They provide a number of drawing shortcuts and enable you to fix drawing mistakes.

This chapter illustrates techniques by which you can manipulate drawing objects. It illustrates how to erase, move, copy, rotate, enlarge and reduce drawing objects. You will also learn how to use the edit functions as a drawing aid tool to make quick corrections.

The chapter ends with a read-through exercise that illustrates the basic steps to draw an engineer's office and an engineering complex plan.

Key Terms in this Chapter

Term	Description
Array	A set of multiple copies of selected objects.
Base point	A reference point used as a guide to move or copy objects.
Editing window	An imaginary rectangle formed by two diagonal points that is used to select drawing objects for editing.
Fillet	Joining lines or arcs at a corner.
Mirror	Creating a new object that is a mirror image of an original object.
Mirror-axis	An imaginary line along which the objects are mirrored.
Move	Moving selected objects from one place to another.
Stretch	Stretching a diagram in a specified direction.
Trim	Cutting drawing objects along a selected edge.

The Role of Edit Functions

The edit module provides great flexibility in changing CADD drawings. If you were to draw using only the drawing functions of CADD, it will probably take the same amount of time as it would on a drawing board. But CADD's editing functions make CADD a dynamic tool that results in significant timesaving.

Changes that may look extremely difficult on a drawing board can be easily accomplished with CADD. Even if you have to make a major change, chances are that you won't have to redraw it from scratch. You can manipulate the diagrams in a number of ways and may be able to rearrange the existing pieces of the drawing to fit the new shape.

The following are the basic capabilities of the edit module (See Fig. 5.1):

- Erasing drawing objects
- Moving drawing objects
- Copying drawing objects
- Changing the appearance of drawing objects

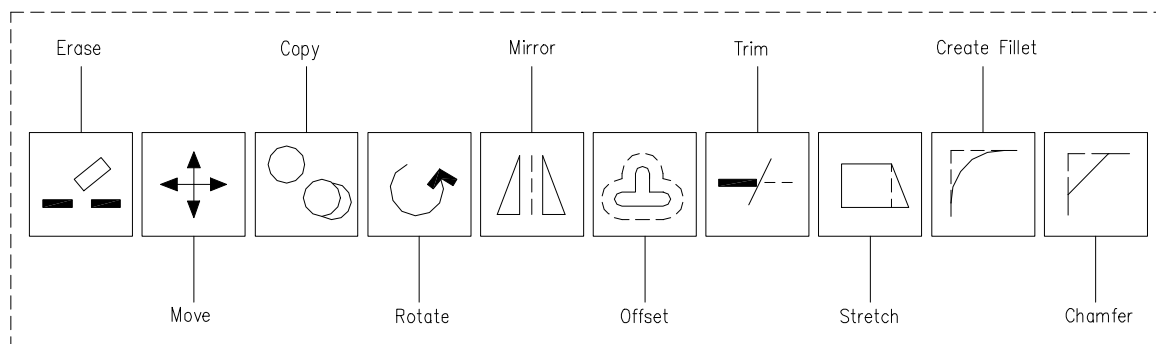


Fig. 5.1: Common examples of editing tool buttons.

Selecting Objects to Edit

In order to apply any of the edit functions, you need to select the drawing objects that need to be edited. CADD provides a number of methods by which you can select drawing objects. You can edit one object at a time, a group of objects, or the entire drawing at once.

*The following are the basic methods for selecting drawing objects for editing:
(The exact procedures vary from one program to another)*

- Selecting objects one by one
- Selecting objects by enclosing them in a window
- Selecting objects using selection filters
- Making groups of selected objects

Selecting Objects One by One

When you need to edit only a few drawing objects, you can select them one by one. To select an object, use the pointing device to position the cursor on the object to be selected and then press the Enter button on the puck or mouse. You can select as many objects as you like by indicating them with the cursor one by one. The selected objects are highlighted with a distinct color or line type.

Selecting Objects by Enclosing them in a Window

When you need to select a number of drawing objects at once, you can select them by enclosing them in an editing window (an imaginary rectangle). You need to enter two diagonal points of the window and all the drawing objects enclosed within that window are selected. There are variations of this function available in many programs. For example, one function selects only the objects that are fully enclosed in the window, while another selects the objects that are just touching the window.

Selecting Objects using Selection Filters

Selection filters let you select objects that meet specified criteria. For example, you can select all drawing objects that are drawn on a specific layer or are drawn with a specific color. Advanced CADD programs even allow you to select similar objects according to their size, shape, line type, etc.

Making Groups of Selected Objects

CADD allows you to form groups of data in a drawing. All the objects selected within a group are edited together when any edit function is applied. The groups are usually formed to contain similar information. This is helpful when you need to edit the same drawing objects again and again. You form a group by selecting objects using any of the methods, and then giving the group a name. When you need to apply an edit function, specify the name of the group, and only the objects within that group are edited.

Erasing Drawing Objects

CADD provides a fast, accurate and convenient means of erasing drawings. Once you have entered the erase command, you are in the erase mode; any objects you select now will be instantly erased. You can use any method described above to select the objects.

Note:

The erase function is very powerful; it must be used with great caution. It can erase large amounts of information within seconds. If you make a mistake while erasing, be sure to take quick action. You can use the undo feature of CADD to disregard the effect of the last command.

Moving Drawing Objects

CADD allows you to move drawing objects within a drawing in a convenient manner. Unlike on a drawing board, you don't need to first erase and then redraw in a new place. You can simply rearrange the existing drawing objects, as you like. This is a very useful tool for analyzing design alternatives and making quick adjustments to drawings.

To move the objects, you need to enter the move function and select the objects to be moved. Enter a base point and a relocation point (destination point) and the selected objects are moved. The base point acts as an anchor for the selected objects and relocation point determines the new location of the selected objects. The objects are moved relative to the base point and the relocation point. Let's say you indicate the relocation point 4'-0" to the right of the base point. All the selected objects will be moved 4'-0" to the right.

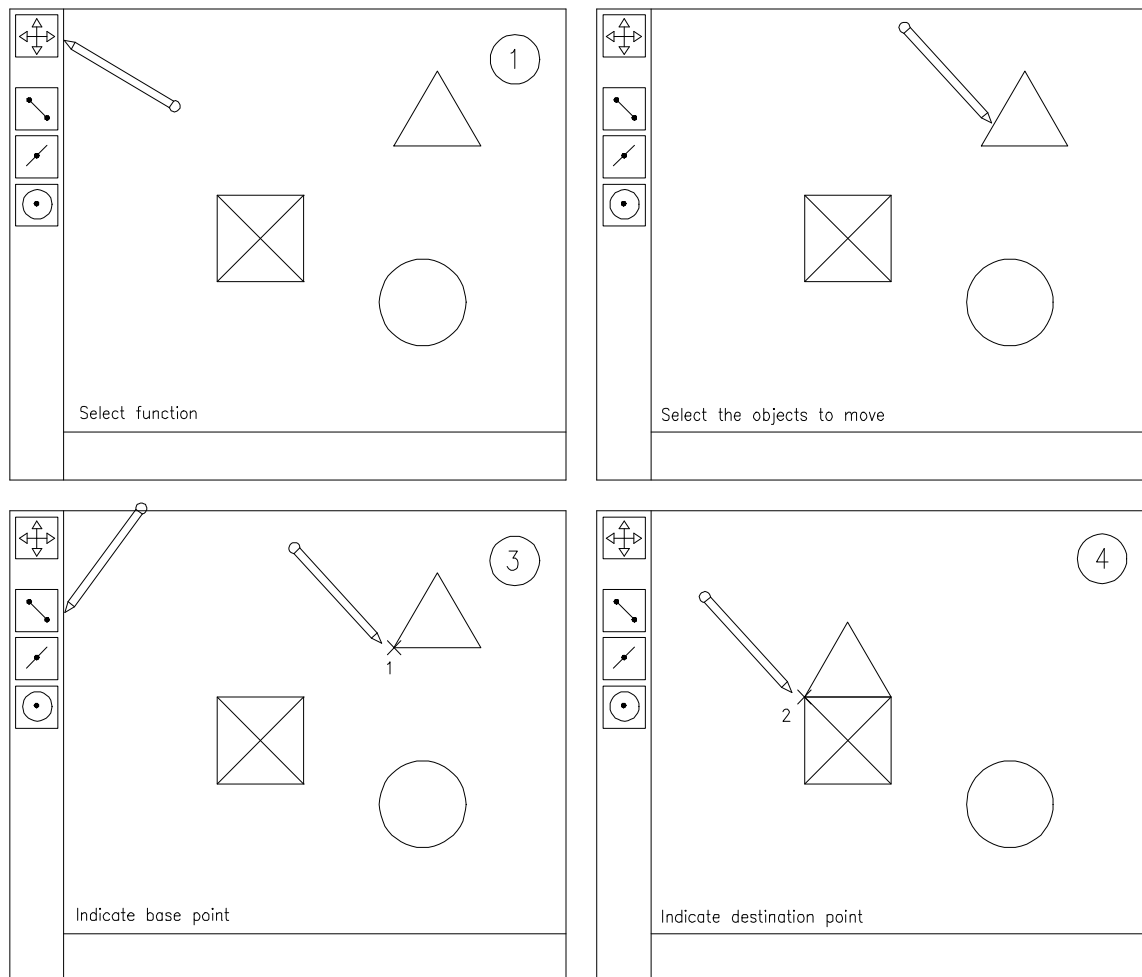


Fig. 5.2: Moving objects using the move function.

Fig. 5.2 (Screen 1) shows a square, a triangle and a circle randomly placed. The following steps illustrate how you can move the triangle on top of the square. For illustration purpose, only the tool buttons used in this exercise are shown.

Step	Action
1	Click on the move tool button. This enters the move function.
2	The computer displays the prompt to select the objects to move. Place the cursor on the triangle and press the Enter button on the pointing device. This should highlight the triangle showing that it has been selected.
3	Screen 3 displays the prompt to enter the base point. First, click on the end point object snap tool button and then place the cursor near the end point of the triangle and press the Enter button on the pointing device.
4	Screen 4 displays the prompt to enter the destination point. Click on the end point object snap tool button again and then place the cursor near the end point of the square and press the Enter button on the pointing device. The triangle is moved exactly over the square as shown.

Copying Drawing Objects

CADD allows you to make quick and easy copies of existing drawing objects. You can copy individual drawing objects or the entire drawing all at once. You can even make multiple copies of drawing objects within seconds.

Using the copy function is quite similar to the way the move function is used. First, you need to select objects using any of the methods described earlier. Then you need to indicate a base point and a relocation (or destination) point. The copied objects are placed according to the relocation point.

Making Multiple Copies in a Rectangular Form

There are separate functions available in CADD that allow you to make multiple copies in a linear or rectangular form (commonly known as a rectangular array). You can make hundreds of copies within seconds. You don't need to enter a base point and a destination point. You just need to select the objects, specify how many rows and columns you need and the distance between them.

Fig. 5.3 illustrates how you can make multiple copies of a chamfered shape. To make five copies of the original shape as shown in Diagram A, you need to specify that the copies be made in two rows and three columns. Enter the distance between the rows as 3/4" and the distance between the columns as 1". Similarly, you can make 44 copies of the original by specifying 5 rows and 9 columns as shown in Diagram B.

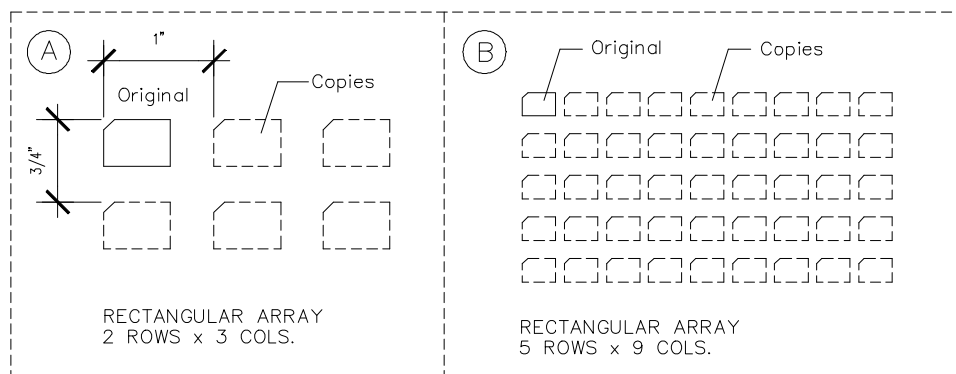


Fig. 5.3: Making multiple copies of a diagram in rectangular form.

Making Copies in a Circular Form

CADD allows you to make multiple copies of selected objects in a circular form (commonly known as a polar array). You need to enter the center point of array, number of copies required and the rotation angle. The center point of the array is a point around which the copies are arranged. The rotation angle determines the extent of the rotation, and the number of copies you specify are equally spaced between the rotation angle.

Fig. 5.4 illustrates how you can make multiple copies of a chair around a table. Diagram A shows the original. In Diagram B, seven copies of the chair are made to fit around half the table (rotation angle 180°). In Diagram C, the same number of chairs fit around three-quarters of the table (rotation angle 270°) and in Diagram D, they fit around the complete table (rotation angle 360°).

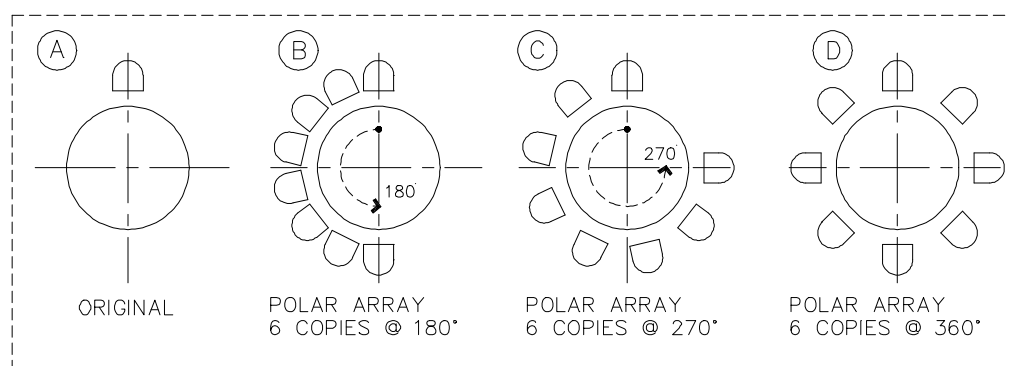


Fig. 5.4: Copying a chair around a table.